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Healthcare AI

AI System Impact Assessment

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Document Control

This document is the responsibility of the Author. It is subject to formal change control after the initial approved release (e.g. version 1.0). Change requests must be communicated to the Author in writing. Agreed changes are recorded in the Version History of the document.

Version History

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Reviewers

Name	Role	Date
Jim Doe	AI Ethics Lead	01/12/20XX

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ISO/IEC 42005 AI Impact Assessment

ISO/IEC 42005 AI Impact Assessment

Introduction

The growing ubiquity of systems, products, services and components incorporating some form of artificial intelligence (AI) has led to an increasing concern about how AI systems will impact society. AI brings with it the promise of great benefits: automation of difficult or dangerous jobs, faster and more accurate analysis of large sets of data, advances in healthcare, etc. However, there are concerns about the potential adverse effects of AI systems, including unwanted bias, environmental harm and unwanted reductions in the workforce.

The development and use of seemingly minor AI systems has the potential to significantly impact (both positively and negatively) individuals, groups of individuals and society as a whole. To foster transparency and trustworthiness of systems using AI technologies, an organization developing and using these technologies can take actions to assure affected interested parties that these impacts have been appropriately considered. No single activity can provide this assurance, but AI system impact assessments play an essential role in the broader ecosystem of governance, risk and conformity assessment activities, which can create a system of trust and accountability.

This AI Impact Assessment has been developed in line with ISO/IEC 42005 to evaluate the impact of the Healthcare AI system provided by <<organization>>.

AI System Impact Assessment for Healthcare AI

In this AI system impact assessment, the potential impacts of Healthcare AI on individuals, groups of individuals and societies are identified, and a determination is made of whether the intended uses of the system are sensitive or restricted and if additional reviews and approvals are necessary.

The Impact Assessment is divided into the following sections:

- Section A – System information: Identification of the AI system that is subject to this AI system impact assessment.
- Section B – Data information and quality: Information about the datasets used with the AI system and the associated data quality characteristics.
- Section C – Algorithms and models information.
- Section D – Deployment environment: Information about the geographical, cultural or legal context of the countries or regions where the AI system is deployed, including information about the technical deployment environment.
- Section E – Relevant interested parties: Information on the relevant interested parties that potentially can be affected by the AI system and its use (stakeholders).
- Section F – Actual and potential benefits and harms: Identify the actual and potential benefits and harms to various stakeholders.
- Section G – AI system failures and misuse or abuse: Document the impact of failures, misuse or abuse on stakeholders.

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Section A – System Information

System Details

Identification of the AI System subject to this AI system impact assessment.

AI system name or identification	Healthcare Data Management and Analysis
Other AI system identifiers	Healthcare AI
AI system lifecycle stage*	Inception, refined throughout development

*AI system lifecycle stage during which this assessment was completed

Revision History

Author	Joel Doe (AI Developer)
Date last modified	01/12/20XX

Approvals

Reviewed by	Jim Doe
Date of review	25/11/20XX
Approved by	Jim Doe
Date of approval	01/12/20XX

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AI System Description

Healthcare Data Management and Analysis (Healthcare AI):
The system enables better patient outcome predictions, research data analysis, and diagnostic assistance.

The system uses deep learning techniques (specifically convolutional neural networks) to image natural language processing and extract information from text data.

Predictive Analytics Engine:
The Predictive Analytics Engine is an AI system that identifies trends, patterns, and potential future occurrences from vast datasets. This is the core engine that Healthcare AI employs, but it is also the basis for other <<organization>> AI applications, including financial forecasting and supply chain predictions.

The system employs regression analysis, neural networks and time-series analysis.

Technologies employed:

- Machine Learning Frameworks: TensorFlow, PyTorch and scikit-learn
- Data Management: Hadoop, Spark, ProvValidator and ProvExplorer
- Cloud Platforms: AWS and Azure for scalable computational resources
- Development Environments: Jupyter, RStudio and MATLAB

See <<reference>> Healthcare AI System Architecture (v1.3, January 20XX), <<reference>> Healthcare AI Requirements Specification (v1.2, March 20XX) and <<reference>> Healthcare AI System Design Specification (v1.6, May 20XX) for further system details.

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AI System Features

	AI System Feature	In current version?	In a planned version? Estimated version and date.
1	Research data analysis	Yes	
2	Diagnostic assistance	Yes	
3	Deep learning (convolutional neural networks)	Yes	
3	Continuous learning	Yes	
3	Interactive query/response system	Yes	
4	Accessible through the online portal, client workstations and mobile devices (IOS and Android supported)	Yes	
5	Management dashboard	Yes	
6	Data consumed - patient data, healthcare professional details	Yes	
7	Feedback on production data for continuous learning	Yes	
8	Cloud platforms for scalable computational resources	Yes	
9	Dependency on Predictive Analytics Engine	Yes	

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Details of Dependencies on Other Systems

The Healthcare AI system uses the Predictive Analytics Engine to provide AI-driven analytics capabilities. The core model processing occurs within the Predictive Analytics Engine. Healthcare AI provides a user interface that 'feeds' the healthcare-related data to the Predictive Analytics Engine and presents the output from the Predictive Analytics Engine to users in a usable format.

The Predictive Analytics Engine models are trained using third-party sourced healthcare datasets. Production data from Healthcare AI is fed back into the Predictive Analytics Engine training data to expand the datasets further.

AI System Purpose

Providing AI solutions for predictive analysis, diagnosis, and healthcare research.

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Intended Uses

Medical device classification: Healthcare AI is classified as Class III as per the EU Medical Device Risk classification.

Medical research and patient diagnoses support: Healthcare AI is intended for use by healthcare professionals within customer organizations, to support research into medical conditions and patient diagnoses. Output is advisory and for interpretation by healthcare professionals, who relate information (such as diagnoses) to patients.

Diagnoses support: Healthcare AI can diagnose a broad spectrum of medical conditions based on third-party sourced medical information, as well as from the medical research that it also supports, thus learning from the healthcare professionals as they undertake research. Healthcare AI can also learn from any modification to its diagnoses by healthcare professionals through continuous learning.

AI subject coverage: Healthcare AI is intended for use for any patient and by any healthcare professional, regardless of ethnic or cultural background, and supports multiple natural languages aligned to the geographical regions in which it is in use.

High availability: The system is available to healthcare professionals 24/7 through various edge devices to support patient diagnoses in customer organization locations (such as workstations in GP surgeries) or the field (using mobile devices or the on-line portal from patient locations).

Access to medical information: Healthcare AI provides a secure front-end to the models within the <<organization>> Predictive Analytics Engine, employing authentication techniques as provided in the edge devices (multi-factor authentication, biometrics, etc.).

Management of medical research: Medical research is intended for use and publication by healthcare professionals, requiring interpretation prior to publication.

Continuous learning: Medical research, patient details and diagnoses information is fed back through continuous learning for use in training, validation and test data.

Regulatory support: Healthcare AI provides transparency in support of medical regulatory requirements and investigations.

Demographics support: Healthcare AI can provide anonymized summary information about medical conditions, such as numbers of patients suffering from a given condition to medical authorities, government agencies or other interested parties.

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Unintended Uses

Use by patients: The system is not intended for use by the general public/patients directly. Patients' unauthorized use of the system could provide 'unfiltered' patient diagnoses directly to patients. Without the interpretation of a healthcare professional, diagnoses could be misunderstood or cause distress to patients.

Human oversight: Healthcare professionals are expected to provide oversight of diagnoses to ensure accuracy and patient confidence.

Unauthorized access to medical information: Unauthorized access to Healthcare AI, or the back-end Predictive Analysis Engine would allow access to sensitive information about patients and their medical conditions/diagnoses, as well as sensitive information about healthcare professionals.

Unauthorized release of medical information: Release of medical information without patient approval, for example to insurance providers, government agencies or to patient employers could have a detrimental impact on the patient's career, wider lifestyle and mental state.

Undermining of established peer review processes: Publishing of medical research directly from Healthcare AI outside of established peer review processes within the medical profession could undermine said processes and patient confidence.

Undermining patient confidentiality: Exploitation of transparency could lead to medical information related to individuals being obtained or inferred.

Automated issuing of prescriptions: Healthcare AI is not intended to issue prescriptions for dispensing to patients without receiving the oversight by healthcare professionals to provide assurance to patients. However, this is seen as a logical next step.

Automated patient diagnoses: A desirable outcome of continuous learning is to refine the model sufficiently so that diagnoses can be delivered directly to patients without needing interpretation by a healthcare professional. This is anticipated to require regulatory approval and some form of multi-jurisdictional certification to be acceptable. As such, this is seen as a long-term goal but is not an intended use at this time.

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Section B - Data Information and Quality

Data Information

Dataset name, version, size	**Medical Database**, v8.5, dated 14/08/20XX Production – 10TB Training – 10TB Validation – 1TB Test - 500GB
Dataset owner	AI Development Team Leader.
Dataset access rights	AI Development Team (R/W) IT Team (R/W) Data Management Team (R/W) Quality Assurance Team (R) Cybersecurity Team (R) Legal (R) Risk and Compliance Team (R) AI Ethics Team (R)
Description of the contents of the dataset	Medical information database. Contains data covering all medical conditions covered by Healthcare AI, designed to be the golden source of all medical knowledge, drawn from third party medical archives and databases. Used by Healthcare AI and the Predictive Analysis Engine to formulate diagnoses based on patient data and consultation data (separate databases), obtained from patients by healthcare professionals.

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	See dataset design documentation for additional information: • Domain details • Data dictionary • Data format details • Data types • Statistics • Data level objectives (DLOs) and data quality objectives (DQOs).
Intended and unintended purposes of this dataset, including if it is approved to be used for AI system	Medical research and patient diagnoses.
Details about data collection, where from, by whom, etc.	Data is sourced from third-party providers under license, including: • PubMed • Cochrane Library • Web of Science • Scopus • SciFinder Web • PsycINFO. Data sourced from providers is subject to the Data Management Team's preprocessing for inclusion in the Medical Database. Data feeds are refreshed regularly to ensure the latest medical information is available. Data handling applied includes: • Exploration • Cleaning • Data analytics • Labelling.

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	Research and diagnosis information from Healthcare AI is continuously fed back into the medical database to enhance the dataset based on real-world applications. This proprietary information is held separately in the database so that third-party sourced data can be refreshed without impacting the proprietary data.
Known risks of unwanted bias in the dataset and geographies covered	The dataset employs (or links to) databases of established medical opinion, which can be publications. By definition, medical opinion can be biased, but this is the nature of the medical profession, addressed through peer review. Notably, these are the same information sources as would be used in 'traditional' diagnosis. Reliance is placed on third-party providers to minimize bias in their databases, i.e., indexing the database to ensure that all relevant data on a specific condition (for example) is considered. Data bias has nevertheless been considered. A broad range of third-party sourced medical information is employed. This represents the breadth of medical knowledge and, as such, does not involve a sample of records (avoiding sampling bias), encompasses all anticipated ethnic and cultural groups (minimizing coverage bias), and does not require a response from individuals to build the dataset (avoiding participation bias). Other forms of statistical biases are not considered to affect the Medical Database. The preprocessing undertaken by the Data Management Team includes labelling, which can result in cognitive or societal biases reflecting the cognitive biases of those performing the labelling. The preprocessing method can cause data-processing bias, where correcting errors, inputting missing information, etc., can be subject to cognitive biases. Completeness of medical records is a significant influencing factor, whereby incomplete medical records for certain groups due to the more fragmented care those groups may receive. This can result in lower-quality medical predictions (diagnoses). Other forms of data bias do not realistically apply to the Medical Database.

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Processes applied to the dataset for quality purposes	Data quality procedures (including data distribution analysis) applied through the Data Management and Quality Assurance Teams are intended to avoid biases (and ensure the reproducibility and robustness of the AI models). Periodic data audits are conducted to ensure there is no introduction of bias in training data. Monitoring by the AI Ethics Team to detect bias in data usage. NOTE that research papers are only included in medical databases once published. There is no direct feed of research papers from the Research Data dataset. Note that data sourced from third parties under contract/license is assumed to be provided with all relevant consents (for example, PII).
Is a data protection impact assessment necessary?	Yes.

Dataset name, version, size	**Patient Records**, v1.1, dated 03/07/20XX Production – 100GB Training – 100GB Validation – 10GB Test – 1GB
Dataset owner	AI Development Team Leader
Dataset access rights	AI Development Team (R/W) IT Team (R/W) Data Management Team (R/W) Quality Assurance Team (R)

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	Cybersecurity Team (R) Legal (R) Risk and Compliance Team (R) AI Ethics Team (R)
Description of the contents of the dataset	Contains patient records, including: • Patient identification and location information • Healthcare professionals involved in treatment • Conditions (diagnosed) • Current diagnoses (for discussion) • Treatments • Medication • Allergies. See dataset design documentation for additional information: • Domain details • Data dictionary • Data format details • Data types • Statistics • DLOs and DQOs.
Intended and unintended purposes of this dataset, including if it is approved to be used for AI system	Medical research and patient diagnoses.

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Details about data collection, where from, by whom, etc.	Patient records obtained from healthcare professionals. Data can be uploaded individually or in bulk as needed. Data sourced from consultations by healthcare professionals regarding patient conditions, symptoms, medications used, etc., are recorded here upon completion of the consultation. Data is subject to the Data Management Team's preprocessing, with the following data handling applied: • Exploration • Cleaning • Data analytics • Enhancement and perturbation • Labelling • Normalization and scaling.
Known risks of unwanted bias in the dataset and geographies covered	Patient records are factual records for the patients being treated by healthcare professionals using Healthcare AI. The data is generally, therefore, not subject to any statistical bias. The labelling undertaken by the Data Management Team can result in cognitive or societal biases reflecting the cognitive biases of those performing the labelling. The preprocessing method can cause data-processing bias, where correcting errors, inputting missing information, etc., can be subject to cognitive biases. Other forms of data bias do not realistically apply to patient records. It is noted that medical records will be based on the opinions of healthcare professionals. Diagnoses by Healthcare AI will be influenced by prior diagnoses in medical records, which could result in incorrect subsequent diagnoses by the AI model.

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Processes applied to the dataset for quality purposes	Periodic data audits are conducted to ensure there is no introduction of bias in training data. Monitoring by the AI Ethics Team to detect bias in data usage.
Is a data protection impact assessment necessary?	Yes.

Dataset name, version, size	**Healthcare Professionals**, v1.2, dated 09/07/20XX Production – 100GB Training – 100GB Validation – 10GB Test - 1GB
Dataset owner	AI Development Team Leader.
Dataset access rights	AI Development Team (R/W) IT Team (R/W) Data Management Team (R/W) Quality Assurance Team (R) Cybersecurity Team (R) Legal (R) Risk and Compliance Team (R) AI Ethics Team (R)
Description of the contents of the dataset	Contains details of all the healthcare professionals using the system. Details include: • Name • Type of practitioner (e.g. doctor, nurse, pharmacist) • Company • Registration number (e.g. NHS number) • Address • Email address

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	• Contact number • Specializations (e.g., GP, surgeon, psychiatrist) • Specialist areas of research. See dataset design documentation for additional information: • Domain details • Data dictionary • Data format details • Data types • Statistics • DLOs and DQOs.
Intended and unintended purposes of this dataset, including if it is approved to be used for the AI system	Records of healthcare professionals using Healthcare AI to support an audit trail of medical research and diagnoses. Linked to patient data dataset.
Details about data collection, where from, by whom, etc.	Entered by healthcare professionals when registering to use Healthcare AI. Data is subject to the Data Management Team's preprocessing, with the following data handling applied: • Exploration • Cleaning • Data analytics • Enhancement and perturbation • Labelling • Normalization and scaling.

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Known risks of unwanted bias in the dataset and geographies covered	No known areas. Data is for reference purposes only, so data bias is not expected.
Processes applied to the dataset for quality purposes	Data entry checks (e.g., use of drop-down lists for type of practitioner).
Is a data protection impact assessment necessary?	Yes.

Dataset name, version, size	**Consultation Data**, v1.5, dated 22/04/20XX Production – 100GB
Dataset owner	AI Development Team Leader.
Dataset access rights	AI Development Team (R/W) IT Team (R/W) Data Management Team (R/W) Quality Assurance Team (R) Cybersecurity Team (R) Legal (R) Risk and Compliance Team (R) AI Ethics Team (R)
Description of the contents of the dataset	Contains transient data collected by healthcare professionals in the treatment of patients, including: • Patient identification and current location information (which may be different from their usual address recorded in patient records) • Healthcare professionals involved in treatment • Current diagnoses (for discussion) • Treatments • Medication.

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	See dataset design documentation for additional information: • Domain details • Data dictionary • Data format details • Data types • Statistics • DLOs and DQOs.
Intended and unintended purposes of this dataset, including if it is approved to be used for the AI system	Patient diagnoses.
Details about data collection, where from, by whom, etc.	Patient records are obtained from healthcare professionals. Data is entered in real time by healthcare professionals regarding patient conditions, symptoms, medications used, etc. Data is stored in the patient records dataset upon completion of a consultation. Data is subject to the Data Management Team's preprocessing, with the following data handling applied: • Exploration • Cleaning • Data analytics • Enhancement and perturbation • Labelling • Normalization and scaling.
Known risks of unwanted bias in the dataset and geographies covered	No known risks. Data is factual and objective, entered by healthcare professionals or from existing medical records. Data bias is not expected. It is noted that medical records will be based on the opinions of healthcare professionals. Diagnoses by Healthcare AI will be influenced by prior diagnoses in medical records, which could result in incorrect subsequent diagnoses by the AI model.

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Processes applied to the dataset for quality purposes	Data entry checks (e.g., use of drop-down lists for medication).
Is a data protection impact assessment necessary?	Yes.

Dataset name, version, size	**Research Data**, v1.5, dated 28/05/20XX. Production – 10TB Training, validation and test data is not required, since this data is not used for model training and development.
Dataset owner	AI Development Team Leader.
Dataset access rights	AI Development Team (R/W) IT Team (R/W) Data Management Team (R/W) Quality Assurance Team (R) Cybersecurity Team (R) Legal (R) Risk and Compliance Team (R) AI Ethics Team (R)

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Description of the contents of the dataset	Contains data recorded by healthcare professionals in medical research. Data is in a free format and can include uploaded documents (such as research papers). See dataset design documentation for additional information: • Domain details • Data dictionary • Data format details • Data types • Statistics • DLOs and DQOs.
Intended and unintended purposes of this dataset, including if it is approved to be used for AI system	Medical research.
Details about data collection, where from, by whom, etc.	Data is entered/uploaded by healthcare professionals. Data is not subject to preprocessing by the Data Management Team.
Known risks of unwanted bias in the dataset and geographies covered.	No known risks. Data is factual and objective, as entered by healthcare professionals. Data bias is not expected. The research materials will inevitably encapsulate the healthcare professionals' opinions (and possible bias). Still, all research is subject to peer review before publication, an established approach which is handled off the system. No unpublished research will be included in the Medical Database.
Processes applied to the dataset for quality purposes	None applied. Research material is subject to peer review which is conducted outside of <<organization>>.

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Is a data protection impact assessment necessary?	Yes. While research papers are generally anonymized before publication, the information gathered during the research process is likely to be patient specific.

Data Quality Documentation

	Data quality characteristics that are met	
	Data quality characteristic	Significance to the AI system, date fulfilled, relevant approvals
1	Accuracy	Does the data in the dataset syntactically and semantically correlate to the intended use of the dataset, i.e., the purpose of the AI system, in this case, healthcare? Accuracy is a required characteristic of Healthcare AI. Fulfilled at: On loading of data and in data quality audits. Approval by: Data Management Team, AI Ethics Team and Quality Assurance Team
2	Completeness	Does the dataset include values for all expected data items? For example, in a dataset describing the locations of all postcodes in the UK, some postcodes, although included, may not have a description of the location. See also representativeness. In healthcare, completeness is difficult to prove since the field of medical knowledge is continuously expanding and being enhanced. This will directly impact the number of edge cases identified and the accuracy of the Healthcare AI model(s). Fulfilled at: On loading of data and in data quality audits. Approval by: Data Management Team, AI Ethics Team and Quality Assurance Team
3	Representativeness	Is the data in the dataset representative of all cases expected to be encountered by the model? To continue the example of completeness, if the dataset is intended to cover all postcodes for Lancashire, it is representative if all postcodes for Lancashire are included but not so if location descriptions are not included. This will directly impact the number of edge cases identified and the accuracy of the Healthcare AI model(s). Fulfilled at: On loading of data and in data quality audits. Approval by: Data Management Team, AI Ethics Team and Quality Assurance Team

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4	Consistency	Is all the data in the dataset coherent and free from contradictions with all other data? Is there another data item contradicting it for any given dataset? This is essential for Healthcare AI since contradictory data could confuse the model, resulting in misdiagnosis and incorrect medication recommendations. Fulfilled at: On loading of data and in data quality audits. Approval by: Data Management Team, AI Ethics Team and Quality Assurance Team
5	Credibility	Is the data in the dataset regarded as believable by the users? In the context of Healthcare AI, users need to feel that credible and reliable sources provide the data processed by the model. Fulfilled at: On loading of data and in data quality audits. Approval by: Data Management Team, AI Ethics Team and Quality Assurance Team
6	Currentness	There can be an inherent ageing of data employed for AI. For example, a financial AI system may employ economic data spanning several decades, which needs to be corrected for inflation, or its interpretation may need to consider shifts in the regulatory landscape. In the context of medical knowledge, this is a continuously evolving field, with old medical research being superseded regularly by new medical research. It is, therefore, necessary to remove out-of-date medical research when replacing it with new medical research on the same subject to ensure diagnoses are based on the latest information. Failure to do so will also impact the consistency of the dataset. Given the changing medical landscape, regarding diagnoses made and research materials, the Healthcare AI and Predictive Analytics Engine must be trained on the most up-to-date medical information. Fulfilled at: During data refresh and in data quality audits. Approval by: Data Management Team, AI Ethics Team and Quality Assurance Team

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7	Accessibility	Refers to the degree to which data can be accessed in a specific use context, particularly by people who need assistive technology or special configuration because of some disability. For Healthcare AI, accessibility is provided through features, predominantly in edge devices, and is not expected to impact the datasets. Fulfilled at: Not required. Approval by: Data Management Team, AI Ethics Team and Quality Assurance Team
8	Compliance	Does the data in the dataset meet regulations, standards, conventions, or other applicable rules? Most data employed by Healthcare AI is sourced from third-party suppliers, which are assumed to meet all required compliance requirements and are covered through contracts/licensing. Production data fed back through continuous learning is assessed for compliance by the Legal and Compliance and AI Ethics Teams. Fulfilled at: On loading of data and in data quality audits. Approval by: Quality Assurance Team, Data Management Team, AI Ethics Team, Legal Team and Risk and Compliance Team
9	Efficiency	The degree to which data can be processed and provide the expected performance levels using the appropriate amounts and types of resources. In the context of Healthcare AI, this relates to ensuring that all medical research is employed and stored most efficiently to minimize storage requirements and reduce processing overheads. This relates directly to the environmental impact of the system. Fulfilled at: At go-live and during performance assessments. Approval by: IT Team, AI Ethics Team and AI Development Team

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10	Precision	Relates to the level of precision employed in numerical data, i.e., the number of decimal points to which numerical data is recorded. In the context of Healthcare AI, to a large degree, this is as defined in the third-party sourced data. Any numerical data fed back from production through continuous learning will be to the level of precision of the information captured or output. Fulfilled at: Not required. Approval by: Data Management Team, AI Ethics Team and Quality Assurance Team
11	Understandability	Assessment: This relates to the ability of users to read and interpret the data using appropriate symbols, units and languages. Data is predominantly sourced from third parties, with production data following the same format as third-party sourced data or easily understandable data such as patient and healthcare professional details. Fulfilled at: On loading of data and in data quality audits. Approval by: Data Management Team, AI Ethics Team and Quality Assurance Team
12	Portability	The ability to move data from one system to another while preserving its quality. Third-party data is sourced under license and is not intended to be available to other systems. Data obtained through continuous learning is proprietary, so it is not intended to be made available externally. However, the datasets are platform agnostic and could be ported to alternative operating platforms if needed. Still, this is unlikely, given that the systems (and models) are coded to operate in the current environments. Fulfilled at: During development. Approval by: AI Development Team, AI Ethics Team

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13	Auditability	Data employed in training, testing, validation, and production needs to be made available for conducting audits. Auditing, especially by third parties, can add to the credibility of the data. In the context of Healthcare AI, this applies to production data fed back through continuous learning since third-party sourced data is expected to have been subject to independent audit under the terms of the contract/license. Data audits are conducted internally by the AI Ethics Team. Fulfilled at: During development and in data quality audits. Approval by: AI Development Team, AI Ethics Team and Quality Assurance Team
14	Identifiability	Can an individual (the PII principal) be identified from the personally identifiable information (PII) data in the dataset? For example, an IP address in a dataset can be used to identify individuals and are considered PII in some jurisdictions and need to be anonymized. In the context of Healthcare AI, many items of PII are contained in the datasets. The nature of the processing performed and the records that need to be maintained render anonymization impossible. Data quality audits are conducted internally by the Quality Assurance Team. Fulfilled at: On loading of data and in data quality audits. Approval by: Data Management Team, AI Ethics Team and Quality Assurance Team
15	Effectiveness	Does a dataset meet the requirements for use in a specific task? This broadly relates to AI uses employed in image processing, i.e., machine vision. In the context of Healthcare AI, the only machine vision employed is facial recognition as part of user authentication, undertaken on edge devices where available, provided by the edge device and outside the context of the Healthcare AI system. Although image data is supplied in the Medical Database, this is not subject to Healthcare AI analysis, but is provided as supporting information as part of diagnoses. Fulfilled at: Not required. Approval by: Data Management Team, AI Ethics Team and Quality Assurance Team

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16	Balance	Refers to the distribution of samples across the different categories of data in the dataset. For a data set containing X number of categories of data items, there needs to be an even number of samples for each data category. While mostly applicable to datasets used for machine vision (see effectiveness), it can also apply in the context of Healthcare AI, whereby the presence of an imbalanced category of a sample population can fail rare instance discovery and classification. For Healthcare AI, balance is addressed by seeking the widest possible sample population from multiple third-party data sources augmented by production data fed back through continuous learning. Fulfilled at: On loading of data and in data quality audits. Approval by: Data Management Team, AI Ethics Team and Quality Assurance Team
17	Diversity	Refers to the difference between samples with the target data. If all or most data records in a dataset are alike, the model trained from that dataset will tend to overfit and be less generalized. In the context of Healthcare AI, the model will tend towards a smaller set of diagnoses, missing rare instances, as with balance. As with balance, diversity is addressed by seeking the widest possible sample population from multiple third-party data sources augmented by production data fed back through continuous learning. Fulfilled at: On loading of data and in data quality audits. Approval by: Data Management Team, AI Ethics Team and Quality Assurance Team
18	Relevance	Refers to the degree to which a dataset is suitable for a given task. In the context of Healthcare AI, this is achieved by using, and only using, medical knowledge sourced from third-party providers augmented with production (medical) data fed back through continuous learning. Fulfilled at: On loading of data and in data quality audits. Approval by: Data Management Team, AI Ethics Team and Quality Assurance Team

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19	Similarity	Essentially, the opposite of diversity refers to the similarity of samples in a dataset in terms of 'interested features'. A model trained on a dataset containing similar samples can have the risk of overfitting similarly to diversity and handled in the same way. Fulfilled at: On loading of data and in data quality audits. Approval by: Data Management Team, AI Ethics Team and Quality Assurance Team
20	Timeliness	Refers to the latency between when an event occurs and when the data is available for use. This primarily relates to tasks on streaming data. In the context of Healthcare AI, timeliness relates to the time it takes for medical knowledge to be made available through third-party sourced data, which lies with the third-party providers, or the time it takes for production data to be fed back through continuous learning. In the latter case, data fed back is subject to preprocessing by the Data Management Team, whose processes must be optimized to minimize latency. Fulfilled at: On loading of data. Approval by: Data Management Team and AI Ethics Team
	Data quality characteristics that are planned but not yet met. Date by which the data quality characteristic is planned to be fulfilled.	
	Data quality characteristic	Significance to the AI system, planned date to be fulfilled
1	None	

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Section C - Algorithms and Models Information

Information on Algorithms

Suitability of the algorithm to the purposes of the AI system and its intended uses	The core AI algorithms are contained in the Predictive Analytics Engine. Healthcare AI provides the user interface and includes additional AI algorithms in this space for imaging and natural language processing for extracting information from text data. Whilst some of the AI algorithms in the Predictive Analytics Engine are generic across multiple fields, including Healthcare AI, some are also tailored to specific uses. Training data specific to healthcare is employed to train models used explicitly in healthcare and for training models in Healthcare AI.
Origin of the algorithm	Algorithms are predominantly based on internally developed, proprietary code, but some use is made of third-party algorithms, and extensive use is made of vendor products in development. Features are determined as part of the requirements analysis and further refined during the design phase. Requirements are subject to approval by management, with the approval process including representation from the AI Development, Data Management, Quality Assurance and AI Ethics Teams. Third-party AI products that can be utilized for specific features are identified as part of the design process. Third-party products are assessed to determine whether their use is cost-effective or if algorithms should be developed internally. Procurement is engaged in acquiring third-party products, as well as the Quality Assurance and AI Ethics Teams. The Data Management Team is also involved in this evaluation approach to ensure it can work with the third-party products. Third-party products are sourced under contract/license, with third parties accountable for the software/data they provide, as long as these are used in line with contract/license terms.

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The scientific validity of the algorithm	Healthcare AI and the Predictive Analytics Engine are proprietary systems that use off-the-shelf third-party products to some extent. The core algorithms are primarily proprietary, with off-the-shelf products used in imaging and natural language processing. Third-party products are sourced under contract/license, relying on third parties for algorithm validation. Proprietary code is subject to testing, validation and verification. The development approach employs Continuous Integration and Continuous Deployment (CI/CD), with the AI Development, Data Management, Cybersecurity and Quality Assurance Teams working closely to deliver quality products, employing: • Both automated and manual testing methods to confirm AI systems' functionality, reliability, and security. • Validation techniques such as back-testing, cross-validation and simulation to ensure the predictive accuracy and reliability of the models. • Controls around training data.
Real-world use and performance	AI system performance is continuously monitored in production through real-time system tracking, emphasizing performance metrics (amongst others), identifying data drift, concept drift and model performance degradation (performance metrics include accuracy, F1 score, AUC-ROC). Production data is fed back into training data for continuous learning. The Data Management Team analyzes and pre-processes this data before it is included in the datasets.
Data requirements	Healthcare AI and the Predictive Analytics Engine use all the datasets listed under section B.
Susceptibility to undesirable outcomes	The following undesirable outcomes have been considered for Healthcare AI (and the elements of the Predictive Analytics Engine employed): • Bias and Fairness: See section B. • Overfitting/underfitting: Possible for all AI models. Continuous learning helps with overfitting and underfitting, ensuring that the models don't just focus on the data they were initially trained with and learn from experience. The <<organization>> approach to continuous learning allows for ongoing refinement of models based on the production data being fed back into the training data. This ensures the model learns and includes the data in future predictions. • Privacy/security: Healthcare AI is inevitably linked to individuals due to its processing of information not only about patients (symptoms, diagnoses, etc.), but also about healthcare professionals using the system.

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	Patients can include children or other vulnerable groups. Patient information can be from healthcare professionals' observations about patients (symptoms). Third-party data is assumed to be anonymized or provided with all relevant consents. The presence of PII within production and training data makes it susceptible to security threats, including unauthorized access and disclosure. • Regulatory and Legal challenges: There are various regulations and legislation relevant to healthcare, such as GDPR, PECR and HIPAA. It is necessary to keep abreast of regulatory and legislative changes that may apply. • Dependency and Reliability: Healthcare AI is intended for use by healthcare professionals, with output for their interpretation before passing to patients. Healthcare AI supports medical research by healthcare professionals, facilitating searches of the Medical Database and employing generative AI techniques to help craft research papers. However, dependency on the system is tempered by the human oversight afforded by healthcare professionals (for medical advice and diagnoses) and peer reviews (for research). • Environmental Impact: Given AI's compute-intensive nature, an inevitable ecological impact is associated with its use. This is offset through cloud processing environments, providing on-demand resources where the cloud providers are contractually bound with sustainability clauses (such as carbon off-set schemes).

Information on Models Used in the AI System

a)	The data (e.g. training, testing, validation) and the algorithms used to develop the models (see 5.5)	See Section B for details of the datasets employed with Healthcare AI. See above re. the algorithms employed (internally developed and externally sourced).

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b)	Model training parameters and methods	Healthcare AI employs linear regression for its less complex functions, such as supporting research. Deep learning techniques (specifically convolutional neural networks) and regression analysis are employed in the core AI code within the Predictive Analytics Engine. The models are generally probabilistic, with Bayesian optimization employed to find the best model parameters. Training techniques employed include supervised learning (with labelling and classification being undertaken as part of preprocessing by the Data Management Team), reinforcement learning, ensemble methods and dimensionality reduction.
c)	Steps should be taken to ensure that data samples are not reused between training and validation or test datasets	Separate copies of the patient records, healthcare professionals, consultation data and research data sets are held for training, testing and validation. As part of data preparation, source data can be either placed into one of the training, testing or validation data sets, or alternatively can be used to generate unique (but related) copies to be placed into each of the data sets. This avoids models remembering specific training cases during testing or validation, but maximizes the breadth of each data set. Given the adoption of continuous learning, transient production data in the consultation data dataset is passed into the patient records dataset and the medical database as necessary, which is then copied into the training, testing and validation copies to enhance the models, as per above.

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d)	Steps taken to train, test and validate multiple models	<<organization>> employs a continuous integration and continuous deployment (CI/CD) approach, through which changes are prototyped in development and subject to testing as part of development before production release. Release into production is through RStudio. The nature of this approach allows the fallback to a prior version for any changed code. This approach is taken across the various models within the Predictive Analytics Engine and Healthcare AI, and involves: • Employing A/B testing strategies to compare new model versions against older ones. • Use of drift detection tools to identify when the data distribution changes significantly, prompting model retraining. • Machine learning optimization techniques
e)	Criteria and steps taken to select appropriate features	Features are determined during inception and further refined throughout design and development. Steps involved in defining and refining features include understanding the dataset and problem, evaluating feature importance, selecting a subset of features and assessing and validating model performance with the chosen features. Specifically, <<organization>>: • Utilize domain expertise to derive valuable features, such as personalized and targeted drug treatments. • Employ feature selection methods, such as recursive feature elimination, to reduce dimensionality. • Use one-hot encoding, binning, or embeddings for categorical data transformation.
f)	Criteria for model selection	The model selection criteria include accuracy, interpretability, complexity, training time, scalability and trade-offs. Probabilistic techniques, including Bayesian optimization, is employed to select the best model parameters.

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g)	Metrics used to evaluate performance of the models (e.g., accuracy, receiver operating characteristic, confusion matrix, root mean squared error or average distance to cluster centre)	Performance metrics include accuracy, F1 score, AUC-ROC.
h)	Evaluation of whether the selected models generalize well on production data	As well as undertaking data preprocessing (Data Management Team), <<organization>> employs validation techniques such as back-testing, cross-validation, and simulation to establish how well models generalize on production data.
i)	Evaluation of whether the selected models perpetuate or create unwanted bias or other harms	There are several measures in place to detect and manage bias in the AI models: • Data quality procedures (including data distribution analysis) applied through the Data Management and Quality Assurance Teams are intended to avoid biases (and ensure the reproducibility and robustness of the AI models). • Periodic data audits are conducted to ensure there is no introduction of bias in training data. • Periodic review and recalibration of AI models to ensure their predictions remain accurate and unbiased. • Monitoring by the AI Ethics Team to detect bias in data usage. • Staff training on the ethical use of AI and the importance of unbiased algorithms. • Use of external AI ethics consultants as required, to provide expertise on eliminating biases.
j)	Evaluation of whether output data from the model contains or creates personally identifiable information (PII)	Yes, Healthcare AI both consumes personal data (e.g., patient details, symptoms, medications taken) and produces personal data (e.g., diagnoses, treatments, medication recommendations).

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k)	Robustness and resilience of the model	Robustness is an essential quality of the <<organization>> AI models and is closely related to privacy and security. <<organization>> considers robustness through data quality procedures (including data distribution analysis) applied through the Data Management and Quality Assurance Teams, risk assessments and quantitative and qualitative metrics recorded for (amongst other things) robustness. Comprehensive model documentation includes robustness scores.
l)	Steps were taken to detect and correct data drift (e.g. retraining the model if the production data starts to differ in characteristics from the original training data)	<<organization>> has a dedicated AI Team for periodic AI model review. Furthermore, <<organization>>: • Continuously monitors models for performance in the production environment • Employs A/B testing strategies to compare new model versions against older ones • Uses drift detection tools to identify when the data distribution changes significantly, prompting model retraining.
m)	Criteria and process for retraining the model	<<organization>> employs continuous learning for the Predictive Analytics Engine and Healthcare AI to enhance model capabilities in operation. Production data fed back through this approach enhances training, testing and validation data so that new model versions can be trained using the latest datasets available to the operational model. This is an integral part of the <<organization>> CI/CD approach, through which AI models are subject to continuous ongoing development, with new versions being released regularly. Retraining the model in operation is not anticipated with this approach, although this will be undertaken if the need is identified.
n)	Impacts of continuous learning (if implemented)	Potential impacts of employing continuous learning for Healthcare AI include introducing bias from opinions rendered by healthcare professionals (which are inevitably opinions based on their experience) and data poisoning attacks (whereby unauthorized access could be exploited to inject erroneous data). The activities of the Data Management Team are designed to address such impacts.

Section D - Deployment Environment

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Geographical Area and Languages

a1	Where the AI system is currently deployed	North America, Europe, Asia
a2	Planned deployment locations	None
b	Legal requirements specific to the geographic deployment area	EU: Artificial Intelligence Act (AI Act) UK: No dedicated AI legislation in the UK. Existing legislation restricts how AI can be used in practice concerning data protection, equality and human rights, and intellectual property. China: Guidelines and legislation have been put in place since 2017. Algorithm provisions around the abuse of algorithmic recommendation systems are in place (since March 2023), including provisions on content management, tagging or labelling, transparency, data protection and fair practices, with additional regulations in certain areas. Japan: Strategies, principles and guidelines published around the major "Society 5.0" project. Canada: The Artificial Intelligence and Data Act (AIDA) aims to regulate international and interprovincial trade in AI systems through risk reduction measures to reduce the risk of harm and biased outcomes. The act provides for public reporting and empowers the Minister to order the disclosure of records related to AI systems. The Act also prohibits certain practices in handling data and AI systems that may cause serious harm to individuals or their interests. The Bill is currently in its second reading in the House of Commons and still needs to be approved by the Senate.

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		India: Some rough draft working papers of the Indian Commission NITI Aayog from 2020, 2021 and 2022 indicate the government's intention to move forward with AI regulation and the National Strategy for AI (2018). The central proposal is introducing a supervisory authority, which, among other things, establishes and administers principles for responsible AI, provides guidelines and standards, or takes over the coordination of authorities in various AI sectors. United States of America: Blueprint for the Development, Use and Deployment of Automated Systems (Blueprint for an AI Bill of Rights). The Blueprint is non-binding and lists five principles to minimize potential harm from AI systems. National Institute of Standards and Technology (NIST) AI Risk Management Framework (in second draft).
c	Cultural considerations of the geographic deployment area	There are no anticipated cultural aspects to consider for healthcare advice. Location can form part of the evaluation process, for example, in providing diagnoses, taking into account conditions more prevalent where the patient is located. Whilst biometrics can be used for authentication when accessing the systems, these are as provided through edge devices, such as mobile phones that can provide fingerprint or facial recognition. <<organization>> does not mandate biometric authentication, but rather as a minimum multi-factor authentication, which is culturally agnostic.

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d	Languages spoken in the geographic deployment area, particularly for natural language processing systems, which are currently supported.	Healthcare AI includes AI for natural language processing for extracting information from text data, including natural language processing of queries entered by healthcare professionals. Algorithms used for natural language recognition are off-the-shelf sourced from third parties under license. Languages are covered for most countries in the regions in which Healthcare AI is available. Healthcare AI is used by healthcare professionals who are assumed to have English as a second language rather than the general public. Medical data (including research papers) drawn from third-party providers are presented in languages supported by Healthcare AI. As per the terms of use of Healthcare AI, any research conducted and uploaded into healthcare must be undertaken in a language supported by Healthcare AI.
e	Behavioural, social and physiological human traits that can be concentrated in geography (e.g., facial expressions, emotional states, gestures or sounds)	See above re. cultural considerations.

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Deployment Environment Complexity and Constraints

How is the AI system deployed or intended to be deployed?	The Predictive Analytics Engine and Healthcare AI core AI algorithms are hosted in scalable cloud computing environments, specifically AWS and Azure. Software and tooling are employed on Wintel (Azure) and Unix (AWS) environments. Healthcare AI can be accessed through: • Online portal: no client-side software is required other than a compatible web browser (all major browsers supported). • PC workstations (including laptops and Apple): client-side software is provided to customer organizations under license. Licenses to access the system are provided on a user basis. There are no restrictions on the number of copies of the client software, and it is incumbent on customer organizations to ensure that new versions of client software are deployed to client workstations as they are issued from time to time. • Mobile devices (iOS and Android): apps are deployed through the App Store and Play Store. <<organization>> developed software using RStudio for both the Azure and AWS environments. Client-side software and mobile apps are packaged for release to customer organizations and to the App Store/Play Store, respectively.

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Constraints to deployment	Client-side minimum requirements are: PC: Windows 7 or later. Apple: Mac OS 10.3 (Panther) or later. iOS: iOS9 or later. Android: Android 10 or later. Note that minimum requirements are subject to change as client-side software is refined over time, which may require operating system features not available on earlier versions.

Section E - Relevant Interested Parties

External Interested Parties

	External relevant interested party name (identify individuals, groups of individuals, vulnerable groups, societies, data subjects, etc.)	Role relative to the AI system
1	Customer organizations	Organizations taking the Healthcare AI service under contract/license. Interested parties within the organizations include: • Procurement • Legal and compliance • IT.
2	Microsoft	Cloud service provider for Azure, scalable computing and storage resources.
3	Amazon	Cloud service provider for AWS, scalable computing and storage resources.
4	ICO	Data protection supervisory authority.
5	HHS Office for Civil Rights	Regulatory body for HIPAA.

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6	Cyberspace Administration of China (CAC)	Lead regulatory body in China.
7	National Institutes of Health	Third-party provider for PubMed.
8	Cochrane Library	Third-party provider for Cochrane Library.
9	Clarivate	Third-party provider for Web of Science.
10	Scopus	Third-party provider for Scopus.
11	Chemical Abstracts Service	Third-party provider for SciFinder Web.
12	American Psychological Association	Third-party provider for PsycINFO.
13	TensorFlow	Provider for TensorFlow, end-to-end open-source machine learning platform.
14	PyTorch	Provider of PyTorch, technical insights and tools.
15	Scikit-learn	Provider of Scikit-learn, tools for predictive data analysis.
16	Apache	Provider of: • Hadoop: an open-source software for reliable, scalable, distributed computing • Spark: a multi-language engine for executing data engineering, data science, and machine learning on single-node machines or clusters.
17	Jupyter	A suite of tools providing software, open standards, and web services for interactive computing across all programming languages
18	Posit	Provider of RStudio, integrated development environment (IDE) for Python.
19	Mathworks	Provider of MATLAB, which combines a desktop environment tuned for iterative analysis and design processes with a programming language directly expressing matrix and array mathematics.

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20	Healthcare professionals	Users of the Healthcare AI system, but also data subjects in their own right.
21	Patients	Data subjects.
22	openprovenance.org	Third party, open provider of ProvValidator and ProvVis, online tools for validating and visualizing data provenance.
23	AI Associates	External legal counsel.

Internal Interested Parties

	Internal relevant interested party name or team name	Role relative to the AI system (e.g. developer, system owner or responsible for oversight)
1	AI Development Team	Responsible for the development and testing of AI models.
2	Data Management Team	Responsible for data processing and management utilizing Hadoop and Spark, across the production, test and validation datasets (see Section B).
3	Quality Assurance Team	Responsible for quality assurance and validation (of models).
4	AI Ethics Team	Responsible for monitoring data usage for biases and ensuring compliance with fairness and transparency guidelines.
5	Cybersecurity Team	Responsible for all aspects of information security throughout <<organization>>.
6	Legal Team	Responsible for ensuring that data management adheres to ethical and legal standards, maintaining relevant regulatory registrations (e.g., with the data protection supervisory authority) and monitoring the evolving regulatory and legislative landscape.
7	Risk and Compliance Team	Ownership and responsible for maintaining the AIMS, performing risk assessments, internal audits and coordinating management reviews.

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8	IT Team	Responsible for the procurement, management and operation of internal IT systems within <<organization>>, including deployment of tools used in developing AI models and Healthcare AI clients to edge devices used internally for access to the system.
9	HR Department	Responsible for the provision of relevant training.
10	CEO	Overall responsible for <<organization>> business globally.
11	Chief AI Officer	Overall responsible for AI development and use within <<organization>>.
12	Finance Department	Enables investments into AI governance systems.
13	Procurement Team	Vets and onboards suppliers providing AI products/services.
14	Employees	Responsible for their appropriate usage of AI, in accordance with the AI Acceptable Use Policy.

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Section F - Actual and Potential Benefits and Harms

	Potential benefits	Potential harms
Accountability	Accountability for AI lies with the Chief AI Officer. This level of commitment provides a high level of assurance to all stakeholders.	Healthcare professionals employ Healthcare AI to assist in patient diagnoses. The advantage of this is to provide more consistent diagnoses of patients with the same conditions regardless of the healthcare professional concerned. However, the output from Healthcare AI is for interpretation by the healthcare professional, who remains accountable for the diagnoses and advice provided to patients. Healthcare professionals have the potential to rely entirely on the output from Healthcare AI rather than interpret output based on their knowledge and experience.
		The Healthcare AI system relies on the accuracy of the medical data sourced into the Medical Database. Accountability for the accuracy of this data lies with the providers. Inaccurate data could lead to incorrect diagnoses.
		Accountability for the healthcare AI system within <<organization>> lies with the Chief AI Officer. The <<organization>> AI Development, Data Management, Quality Assurance, AI Ethics, Legal, and Risk and Compliance teams are highly experienced and provided with regular training on AI-related matters. Accountability for training sits with HR. The use of personnel that do not possess the

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		necessary capabilities could result in inaccurate models, introduction of bias, etc.
		Healthcare professionals are accountable for ensuring patient and consultation data are entered correctly. Patients cannot modify this data themselves but can request correction of errors through their healthcare professional. Healthcare professionals are accountable for the accuracy of the information held within the Healthcare Professionals dataset.
Transparency	Transparency can give healthcare professionals confidence in the output being provided to them. Transparency is a topic communicated through: • Terms of use (reviewed by legal and compliance) • Educational videos on the products and services offered provide summary information about the capabilities of the systems and the management dashboard. • Whitepapers on various topics related to AI, available on the <<organization>> website or through customer support channels • Capabilities are set out in terms of use within contracts and licensing agreements, with aspects articulated during user registration, at logon and accompanying output. Note that healthcare professionals use Healthcare AI under contract/license. • Statements at logon and accompanying output that the system is used for decision-making (diagnoses).	<<organization>> does not volunteer information regarding the algorithmic logic, traceability, auditability or testing methods used for Healthcare AI as part of the normal output from the system. This is intended so as not to swamp healthcare professionals with information that may cloud their desired output. However, this may undermine the healthcare professional's confidence in the output. For this reason, Healthcare AI provides a management dashboard for select users within customer organizations, to monitor the performance of the AI, with further information available through customer support channels. Such users are granted access to the management console on request by customer organizations. Note that it is incumbent on customer organizations to ensure appropriate approval of such access, requests for which will be processed in good faith by the Customer Service Team when they are submitted through established customer contacts.

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	Results (output) are provided in a human-readable format that can be easily understood by the intended audience (healthcare professionals).	Transparency can indicate that further consideration may be required if the healthcare professional considers the model unreliable or the output is not based on suitable information.
	Transparency is a core element of the <<organization>> risk assessment, development, and data management approaches: • Models, training data and other assets are evaluated for transparency (amongst other things) as part of the risk assessment of assets (in the risk assessment template). • Development considers topics such as the ability to monitor the system, metrics and reporting capabilities. • The Data Management Team considers transparency of the data sourced from third-party sources, and that of production data fed back for continuous learning. All these allow for identifying areas where a lack of transparency is being provided. This is particularly beneficial in dealing with third parties, whereby <<organization>> can re-evaluate the use of a third party that cannot demonstrate adequate transparency.	Healthcare does not routinely provide information on the datasets and their sources for similar reasons. However, this information is again available through customer support channels, through which queries about the results provided can also be made.

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	<<organization>> employ off the shelf tools providing transparency capabilities (TensorFlow, Scikit-learn), model insights to help provide transparency (PyTorch), use of open-source tools for transparency in the code employed (TensorFlow, Hadoop, Spark) and integrated development environments for code management (Jupyter, RStudio). Assessments for transparency (as well as other metrics) can be made available to customers.	
Fairness and discrimination	Demonstrable elimination of bias in Healthcare AI underpins reliability and confidence in the output from the system.	Bias in the output from Healthcare AI could result in an incorrect diagnosis, treatment, or choice of medication. Given that Healthcare AI is intended for use by healthcare professionals, there is a reliance on the interpretation of the output to eliminate the effect of any bias. However, healthcare professionals could inject bias into the process, for example, into a preferred medication. It should be noted, however, that such bias may be inherent in any interaction with a healthcare professional, whether supported by Healthcare AI or not.
	Diagnoses from Healthcare AI are based on information provided by healthcare professionals and medical facts provided through the Medical Database. The susceptibility of certain groups (ethnic, gender, regional, smokers) to different conditions could be perceived as bias in the source data. Still, it is beneficial because it allows more consistent diagnosis once a patient's particular grouping (or groupings) is established.	In providing diagnoses about individuals, information produced and stored by Healthcare AI could place an individual at a disadvantage, should it become more widely known, for example impacting their employment prospects or ability to seek life insurance, etc.

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	Healthcare AI is voluntary, provided under contract/license to organizations with potential fairness and discrimination issues set out in terms of use.	Using the same data for training, testing, and validation can result in bias within training data (and the model) from being undetected since testing and verification will yield the same results, so variations in output from the model from different inputs will not become apparent.
		Edge cases indicate model degradation, which can manifest in unwanted bias in model output. Continuous monitoring is undertaken in production through real-time system tracking, emphasizing performance metrics (amongst others), identifying data drift and model performance degradation to help reduce bias.
	There are several measures in place to detect and manage bias in the AI models: • Model bias is evaluated under the risk category of 'fairness' through the risk assessment process applied to assets (in the risk assessment template). • Data quality procedures (including data distribution analysis) applied through the Data Management and Quality Assurance Teams intended to avoid biases (and ensure reproducibility and robustness of the AI models). • Model training techniques include supervised learning (with labelling and classification being undertaken as part of preprocessing by the Data Management Team), reinforcement learning, ensemble methods and dimensionality reduction.	Use of poor-quality data, i.e., data with poor accuracy, completeness, veracity (credibility), relevance, integrity or usability, or data that is out of date, can result in biased and unfair output or possible misdiagnosis, which other than the obvious health implications could also result in discrimination against a patient. To help address this: • Data is sourced from reputable third parties and subject to management and preprocessing by the Data Management Team. Tools and techniques include the use of Utilize ETL (Extract, Transform, Load) pipelines to cleanse and prepare data, outlier detection using methods such as the Interquartile Range (IQR) technique and the use of normalization and standardization techniques to ensure data consistency. • Completeness is subject to ongoing improvement through continuous learning, whereby production data is fed back to expand the training data set. Production

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	• Rigorous testing of the AI system, both before use and periodically afterwards, to ensure no disparate impact on a protected class of individuals? • Periodic data audits are conducted to ensure no bias in training data is introduced. Fairness auditing tools are employed to assess demographic parity (model predictions are consistent across all demographic groups) and equalized odds (measure fairness by comparing false positive/negative rates). • Periodic review and recalibration of AI models to ensure their predictions remain accurate and unbiased. • Monitoring by the AI Ethics Team to detect bias in data usage. • Staff training on the ethical use of AI and the importance of unbiased algorithms. • External AI ethics consultants are used as required to provide expertise on eliminating biases.	data is fed into the training data by the Data Management Team, which applies preprocessing techniques for third-party-sourced data. • The Data Management Team works with the AI Development and Quality Assurance teams to ensure the datasets' relevance, integrity and usability. Data sourced from third parties is assumed to be provided with all relevant consents (e.g., for using PII). As part of the terms and conditions in customer contracts, <<organization>> reserves the right to continuously use production data to improve products (enhancing training data). Terms and conditions of employment at <<organization>> include a clause to use employee data for ongoing improvement of products. Customer contracts and terms and conditions of employment are subject to review by Legal and Compliance.

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Privacy	Production data fed back through continuous learning can be from observations made about patients (symptoms) from healthcare professionals. The healthcare professional (as an individual, per above) enters data into the system. It is incumbent upon healthcare professionals to inform patients how their information is used.	Healthcare AI processes PII, including sensitive medical records. Such data is both consumed (input by healthcare professionals) and produced in the form of diagnoses and medication requirements. All this data, including details of medical conditions and location information, are held in the Patient Records dataset. Personal information about healthcare professionals, including employer and location information, is in the Healthcare Professionals dataset. Unauthorized access to or disclosure of such information would contravene privacy legislation, such as GDPR and HIPAA.
	Handling of sensitive data is subject to controls within the Data Management, AI Development, Quality Assurance and Cybersecurity teams, as well as controls within IT and cloud service providers. This collaborative effort helps address privacy issues within healthcare AI.	Disclosure of medical records could impact patient confidentiality, resulting in discrimination or unfair treatment by employers or the general public. Such could impact patient careers and likely impact patient mental health.

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	Given the privacy implications of the data held, <<organization>> is registered with the ICO to process personal data and has implemented a Privacy Policy. Registration and compliance activities with this and other regulatory bodies are managed through the Risk and Compliance Team. Further, <<organization>> has implemented a cybersecurity program, managed through the Cybersecurity Team, with a defined Information Security Policy and information security processes and controls pervasive throughout the organization, as well has contractual requirements with third parties, such as cloud service providers. Though driven by business needs, the above also benefits <<organization>> employees.	Privacy legislation (e.g., GDPR) restricts the transfer of PII across national boundaries to jurisdictions that do not have suitable data protection legislation. If data taken from a patient in an overseas location is transferred across national borders in contravention of this legislation, the individual's data could be at higher risk of disclosure. Data protection legislation is in place in locations in which <<organization>> operates, including: • Canada (PIPEDA) • Japan • Jersey • Guernsey • Switzerland • Isle of Man These locations are considered 'safe' for data protection in the UK. Healthcare AI data is held in scalable computing resources with cloud service providers, with whom <<organization>> has arranged for data to be stored in data centres within UK national boundaries.
	The principles of necessity, proportionality and data minimization are adopted to minimize exposure from PII. However, note that the broadest possible dataset for model training will allow the model to be fine-tuned for accuracy.	Datasets sourced from third parties under contract/license are assumed to be provided with all relevant consents (for example, PII). Where this is not the case, privacy issues may arise.
	The Legal Team identifies retention periods for various data types. A retention policy has been defined. It should be noted that training/verification/validation data	<<organization>> has a direct relationship with healthcare professionals as members of customer organizations. This relationship is managed through customer contracts/licenses. There is no direct relationship with data subjects (patients), so

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	and production data used to enhance such datasets through continuous learning are not subject to retention periods other than the lifetime of the system concerned. The reason for this is that by removing data that has exceeded an arbitrary retention period from a training dataset could have unpredictable results for the AI model. Terms of use in contracts set out how production data fed back for continuous learning will be used.	<<organization>> relies on healthcare professionals for this relationship.
		Use of PII for purposes outside of Healthcare AI, such as marketing activities, could further expose the data to misuse and would likely contravene ICO registration. <<organization>> does not use PII collected for Healthcare AI outside of the system.
		Given that healthcare professionals use Healthcare AI through customer organizations, <<organization>> has no direct relationship with patients (data subjects). It is incumbent on customer organizations to manage this.
Reliability	Demonstrable reliability of Healthcare AI underpins patient and healthcare professional confidence in the system. Confidence in Healthcare AI provides <<organization>> with a competitive advantage.	Given Healthcare AI provides patient diagnoses and medication requirements to healthcare professionals, reliability issues with the system could manifest as incorrect diagnoses and medication recommendations. Whilst healthcare professionals are expected to apply their professional judgment, reliance on the system could result in misdiagnoses or incorrect medications going undetected.
	Regulatory requirements that impact reliability include GDPR, PECR and HIPAA, and compliance is handled through the Risk and Compliance Team, working with other teams to ensure a quality product. This collaboration provides the desired demonstrable reliability.	Misdiagnoses could be an identification of a more severe complaint than exists, which could cause undue distress and psychiatric harm.

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	Security and other updates are carried out through maintenance schedules agreed upon with customers. Updates are made through change management. Customers are advised of the change window when an emergency change needs to be applied immediately. Release notes include a record of the changes made in an update. These measures are intended to provide customers with confidence in the system's reliability.	To address these potential impacts, <<organization>> employs Continuous Integration and Continuous Deployment (CI/CD), with the AI Development, Data Management, Cybersecurity, AI Ethics and Quality Assurance teams working closely to deliver quality products, employing: • Both automated and manual testing methods are used to determine AI systems' functionality, reliability and security. • Validation techniques such as back-testing, cross-validation and simulation to ensure the predictive accuracy and reliability of AI models. • Controls around training data. AI system performance is continuously monitored in production through real-time system tracking, emphasizing performance metrics (amongst others), identifying data drift and model performance degradation (performance metrics include accuracy, F1 score, AUC-ROC). Generally, monitoring is analytical and exception-based, whilst audits are static 'point-in-time' checks to detect failures and variations from expected performance. Any failures or variations identified are, of course be addressed.
	Reliability is covered as part of terms and conditions, but reliability information is also available through management dashboards and, where appropriate, accompanying output.	Incorrect medication or dosage could cause risk to life.

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	Testability and auditability are facilitated through the transparency measures set out above, but also in the quantitative and qualitative metrics recorded for performance (precision, recall, F1 Score, AUC-ROC), transparency, reproducibility, robustness, fairness (breaches), model evaluation , e.g., Mean Absolute Percentage Error (MAPE) or Root Mean Squared Error (RMSE) and computational costs, as well as the reporting around these metrics, all of which is available to demonstrate reliability to customers.	Vulnerabilities in developed software or software sourced from third parties can impact the reliability of the Healthcare AI system. Vulnerability management is part of <<organization's>> development approach (which includes code reviews and the use of static vulnerability scanning tools). Platforms used in the delivery of the systems are located in the cloud. Vulnerabilities on such platforms could equally have an impact on system reliability. Scanning of these platforms sits with the cloud service providers as set out in terms and conditions. Customers can report vulnerabilities through customer support channels, and internal users can report vulnerabilities through the IT Help Desk.
	System errors are handled on a break-fix basis, whereby detected mistakes or failures are investigated and managed through incident management and, as necessary, change management (depending on the nature of the failure and required fix). Catastrophic system failures are managed through disaster recovery plans (within <<organization>> and cloud service providers) that are subject to testing. Edge devices are operated by customers, for which environments customers are assumed to define their disaster recovery plans, but <<organization>> does not mandate this.	Misdiagnoses could include a failure to identify serious conditions, resulting in life risk.

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	<<organization>> employs a Continuous Integration and Continuous Deployment (CI/CD) approach, through which changes are prototyped in development and subject to testing. Implementing CI/CD pipelines to facilitate automated testing and seamless deployments of AI models helps ensure consistency and reliability in solutions delivered. Release into production is managed within RStudio. The nature of this approach allows easy and rapid fallback to a prior version for any changed code.	
Safety	Healthcare AI does not provide for any mechanical actuation that could cause harm to a user or a third party. Whilst risk to life is a possibility, this relates more closely to reliability in the context of Healthcare HR, which has been covered above.	
Explainability	Explainability is closely related to transparency in how it is addressed (see above) and helps provide trust in the output (and, therefore, the Healthcare AI system).	A lack of sufficient information in the output from Healthcare AI can impact the ability of healthcare professionals to make an informed decision. Whilst healthcare professionals are expected to apply their professional judgment, this is only possible if the output provides sufficient information.
		Healthcare professionals can take up issues with the output from Healthcare AI through the contracting customer organization.
Environmental Impact	Sustainability is a core value of <<organization>>, which involves developing systems considering environmental impact. This provides a competitive advantage for <<organization>> and benefits society as a whole.	The environmental impact of AI technology centres around power consumption and heating/cooling for compute-intensive environments. Given AI's compute-intensive nature, this is unavoidable, although <<organization>> continually strives to improve algorithmic efficiency to reduce operational costs and environmental impact.

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	AI technology is inherently compute-intensive, although such risks are transferred through cloud processing environments, providing on-demand resources where the cloud providers are contractually bound with sustainability clauses.	Failure to consider environmental impact undermines <<organization's>> reputation and position within the marketplace.

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Section G - AI System Failures and Misuse or Abuse

Impact of AI System Failure

	Failure description	Impacts resulting from potential failure (include if it is a benefit or impact)
1	Inaccurate input data (data entry by healthcare professionals)	Healthcare AI depends on the information healthcare professionals provide to render accurate diagnoses. Inaccuracies in input data could result in inaccurate diagnoses and medication recommendations. In extreme cases, this could cause risk to life and is likely to, at a minimum, cause patient distress.
		Inaccurate input data could impact model training since data is fed back through continuous learning into training, validation and test data. This can result in model degradation, impacting future diagnoses.
2	Poor model training	The Healthcare AI models are trained using information in the medical database and employ data fed back through continuous learning. The Medical Database is primarily sourced from third parties and augmented from the production data fed back. Errors in the dataset will impact the AI model(s) training and the accuracy of diagnoses/medication recommendations.
3	System failure - server failure	Healthcare AI is intended to provide 24/7 support to healthcare professionals in any location through various edge devices. Server failures causing a system-wide failure would impact the availability of the system. Whilst this would be detrimental, the Healthcare AI service is advisory, with healthcare professionals still able to provide patient care in its absence.
4	System failure - process/component failure	Healthcare AI and the Predictive Analytics Engine on which it draws are complex systems operated across multiple physical and virtual servers in cloud environments, in which numerous processes and data stores are employed. All components/processes are required to allow Healthcare AI to provide the correct output. Though failure of a component/process is likely to result in a system-wide failure, the system architecture includes components/processes that regulate the system's operation or ensure correct recording of production data. Failure of such components would not necessarily cause a system-wide failure but rather could result in erroneous output or the inability to process production data being fed back for continuous learning.
5	Data loss	Healthcare AI is highly dependent on the source data held in the Medical Database and Patient Records datasets and, to a lesser degree, on the Healthcare Professionals and Consultation Data datasets. Each of

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		these datasets exists in relational databases spread across multiple database files. Any of these individual data stores could be subject to corruption or data loss. The loss of an entire dataset will likely cause a system failure, impacting system availability. However, the partial loss or corruption of data within a dataset in use by an operational AI model may have unpredictable results, leading to possible misdiagnoses or incorrect medication recommendations.
6	Edge cases	Every application of AI inevitably encounters scenarios where the system does not perform as required or expected, making unexpected correlations or drawing incorrect conclusions. Such cases are termed “edge cases". They are usually caused by training the AI model using insufficient training data, such that the model is provided input data scenarios that it does not recognize. Generally, the broader the training dataset, the fewer edge cases should occur. In the context of Healthcare AI, this could result in misdiagnoses and incorrect medication recommendations.
7	Poor system performance	Edge cases are an example of poor system performance, but there are many other manifestations, such as bias and unfair decision-making. These can impact the accuracy of output from Healthcare AI, with associated potential for discrimination, patient distress, mental health impact and risk to life as set out above and in Section F.

Impact of AI System Misuse or Abuse

	Misuse or abuse description	Impacts resulting from potential misuse or abuse (include if it is a benefit or impact)
1	Unauthorized access	As Section F states, unauthorized access to information within or produced by Healthcare AI could compromise patients' and healthcare professionals' privacy.
		As well as disclosure of information, unauthorized access could result in corruption or intentional modification of data on which Healthcare AI depends. Consequences are consistent with those set out for data loss above.

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		Unauthorized access can undermine confidence in the system for customers (including healthcare professionals), patients and the wider community. This can have a detrimental effect on <<organization's>> reputation, have a material impact on <<organization's>> ability to secure financial backing and incur financial penalties from regulators, such as the ICO, all of which can affect <<organization's>> operating as an ongoing concern.
2	Intentional erroneous data input	It is conceivable that authorized users could intentionally enter erroneous or incomplete data to Healthcare AI (perhaps healthcare professionals with a grudge or disgruntled employees). The impact of such would be consistent with that for inaccurate input data above.
3	Vulnerability exploitation	Poor management of system vulnerabilities (either in the underlying infrastructure, vendor-sourced products or within developed software) can result in unauthorized access to Healthcare AI, with the impacts as set out for unauthorized access set out above or potential full or partial system failures, as set out for system failures above.
4	Malware	Introducing malware to Healthcare AI or the Predictive Analytics Engine could impact the system's availability or the accuracy of output from the system, as set out for the system failures above.

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